

EE 3202.304

ECE Fundamentals Lab II

Lab 2: Transformers

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I. Experemental Results

Part 1

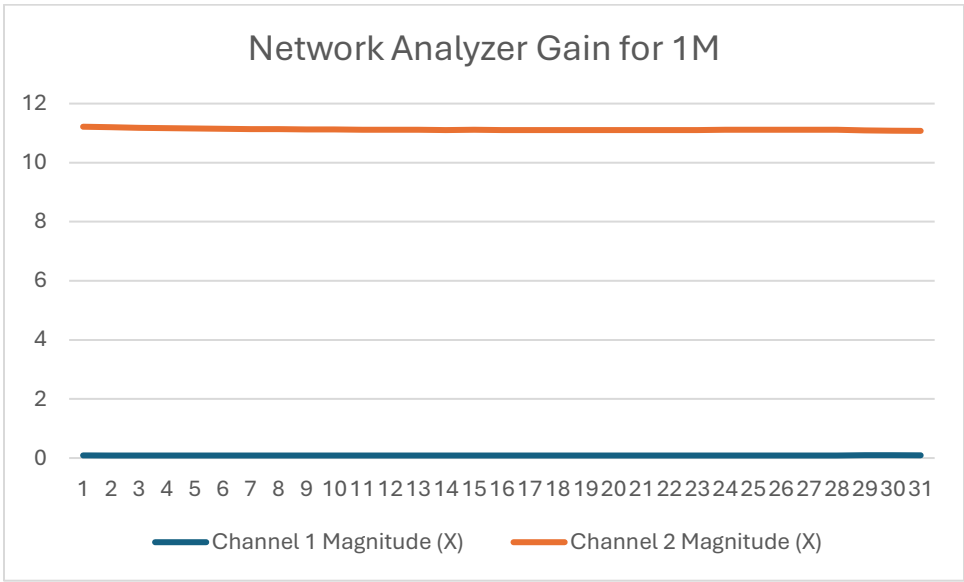


Figure 1.1: Primary vs Secondary ratio with 1M ohm resistor.

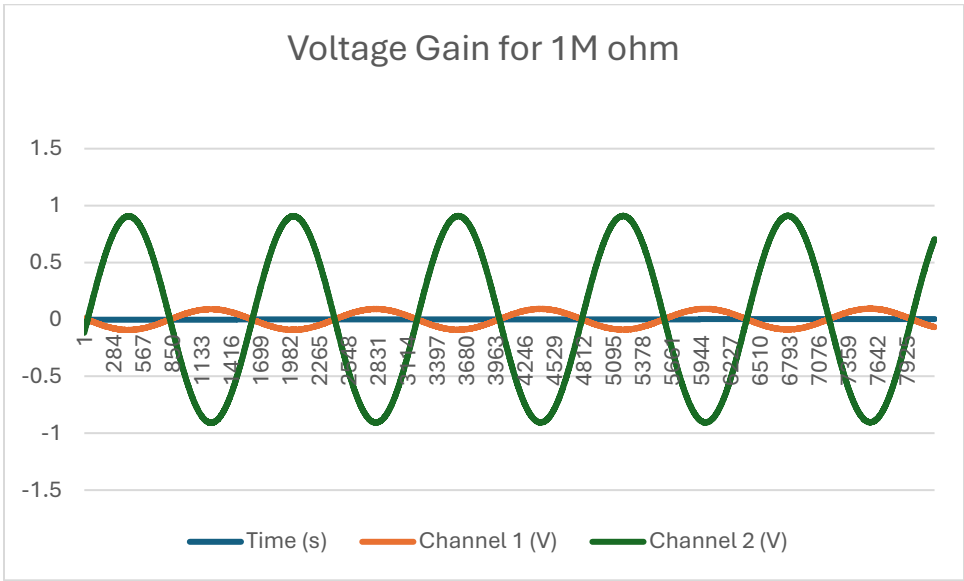


Figure 1.2: Primary vs Secondary voltage gain with 1M ohm resistor.

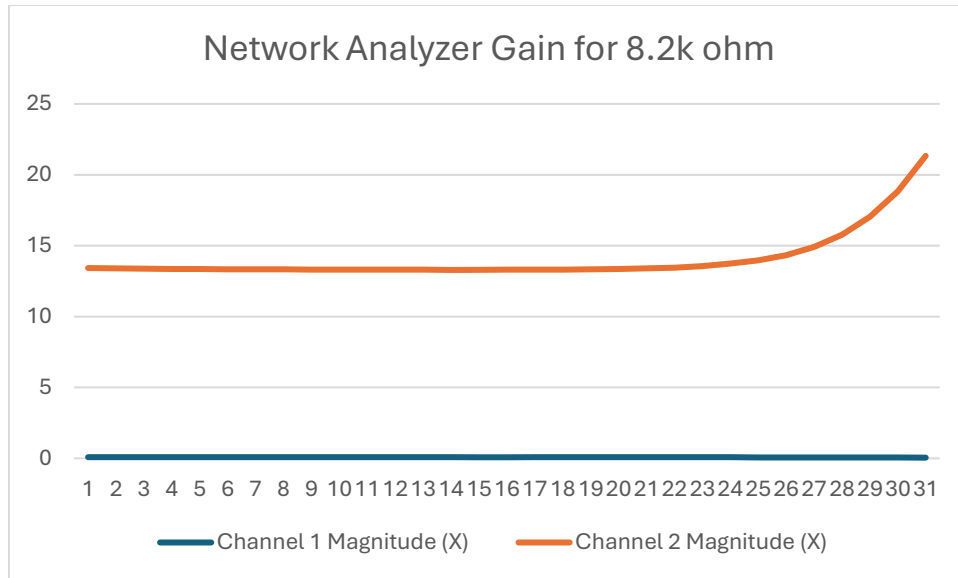


Figure 1.3: Primary vs Secondary ratio with 8.2k ohm resistor.

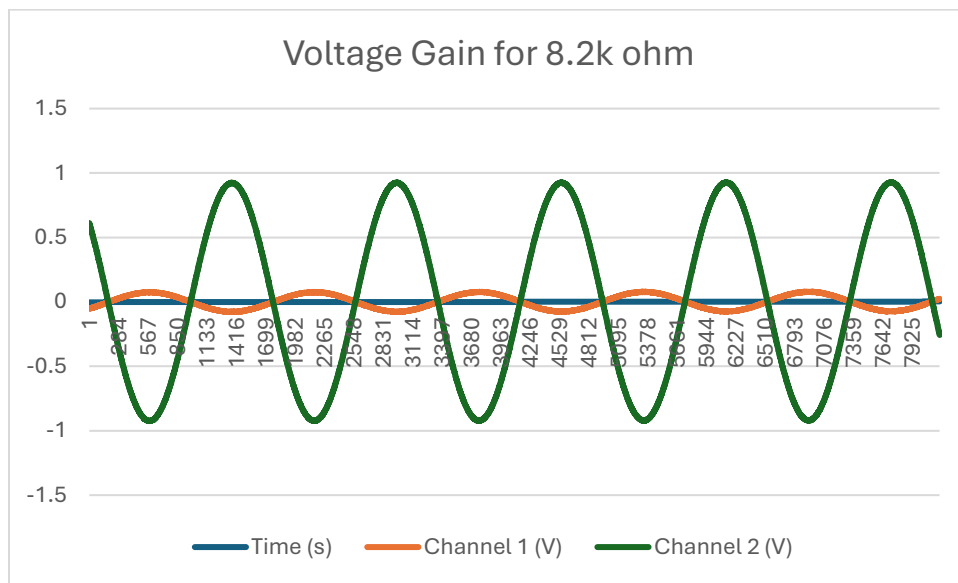


Figure 1.4: Primary vs Secondary voltage gain with 8.2k ohm resistor.

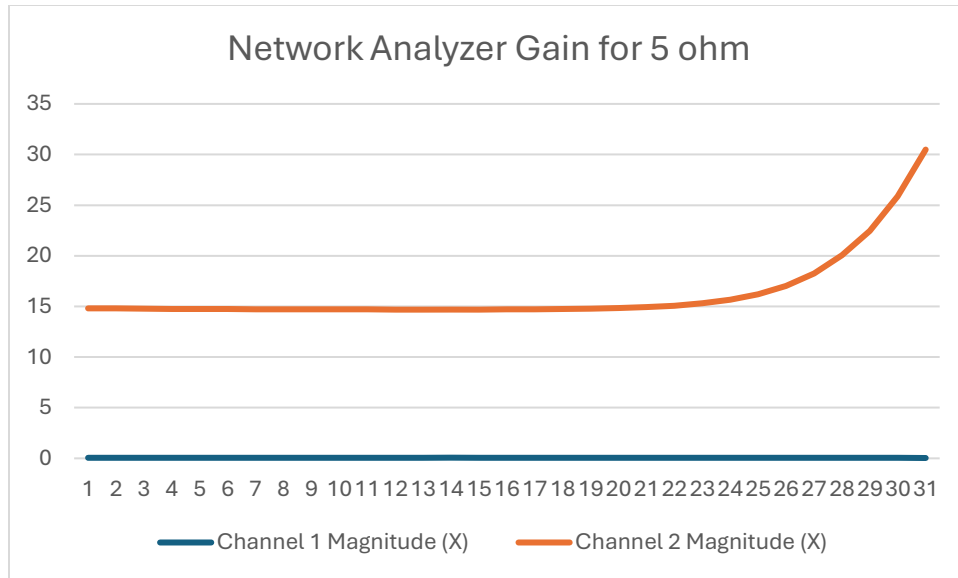


Figure 1.5: Primary vs Secondary ratio with 5-ohm resistor.

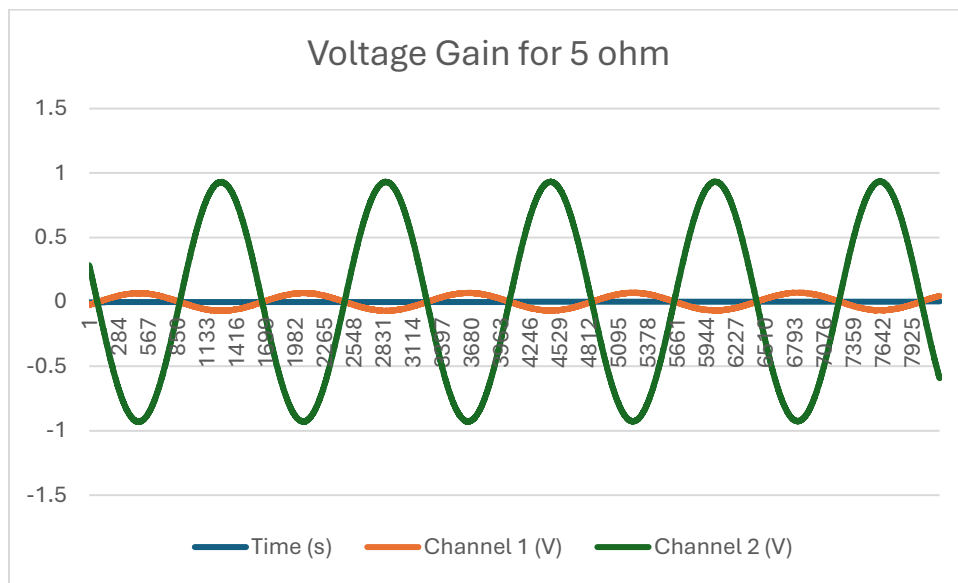


Figure 1.6: Primary vs Secondary voltage gain with 5-ohm resistor.

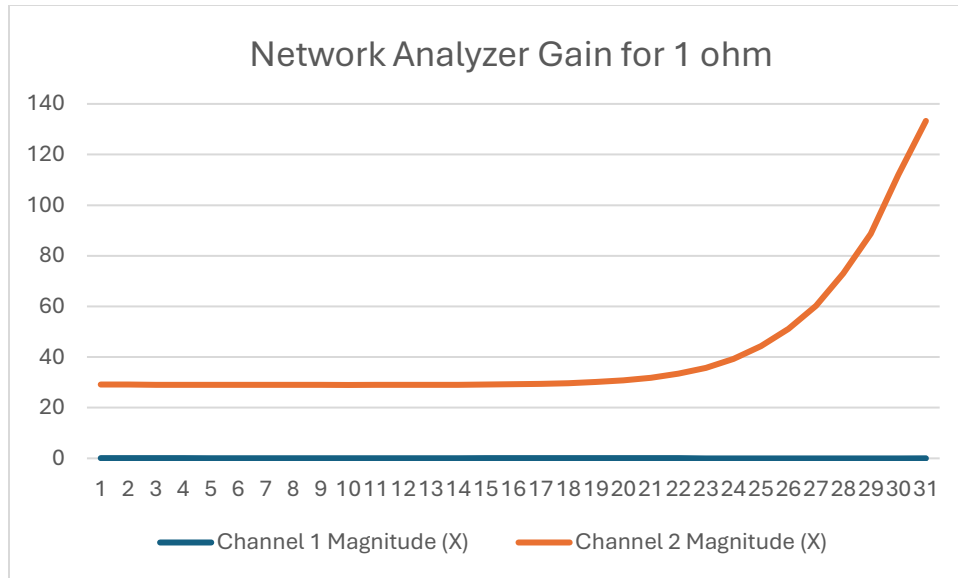


Figure 1.7: Primary vs Secondary ratio with 1-ohm resistor.

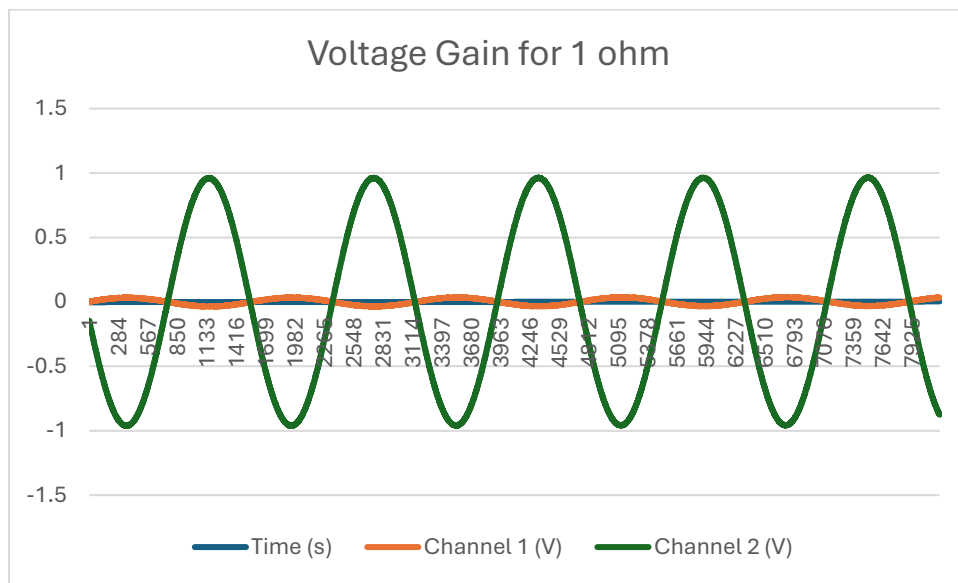


Figure 1.8: Primary vs Secondary voltage gain with 1-ohm resistor.

Part 2

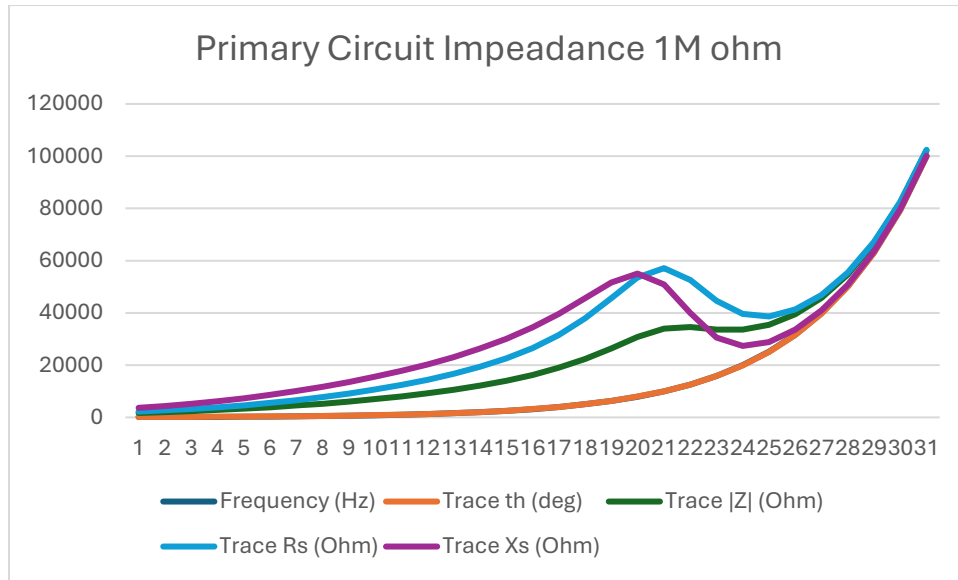


Figure 2.1: Primary Side Impedance for 1M ohm resistor.

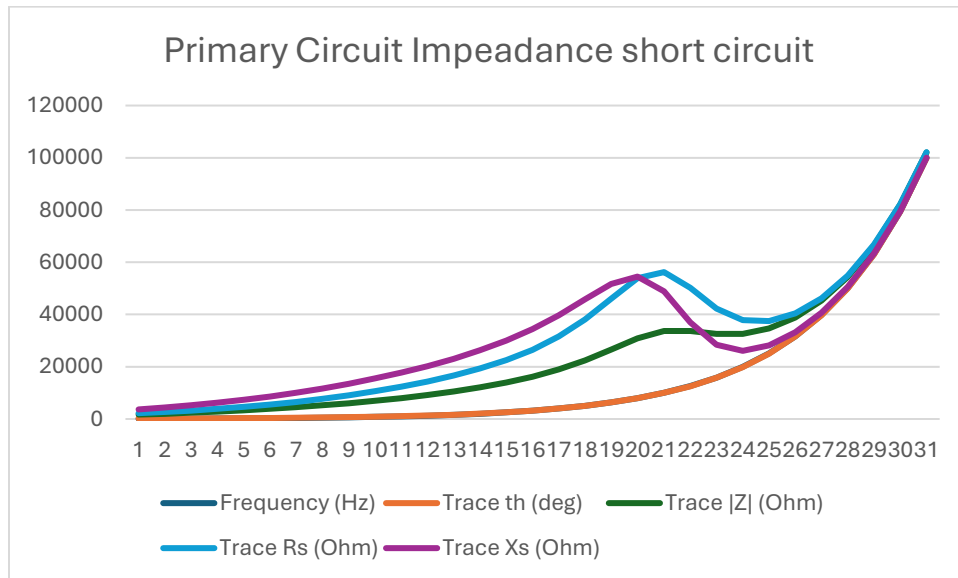


Figure 2.2: Primary Side Impedance for short circuit.

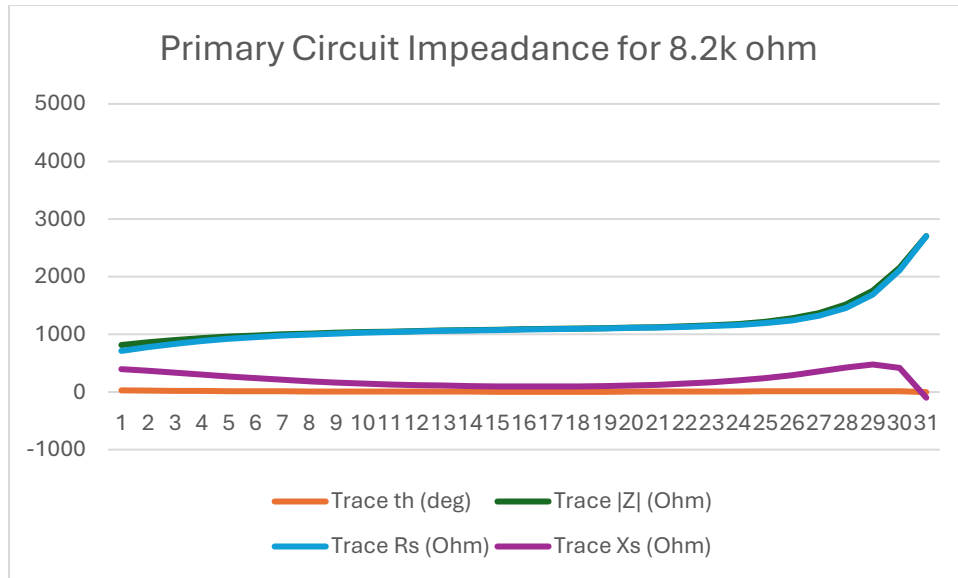


Figure 2.3: Primary Side Impedance for 8.2k ohm resistor.

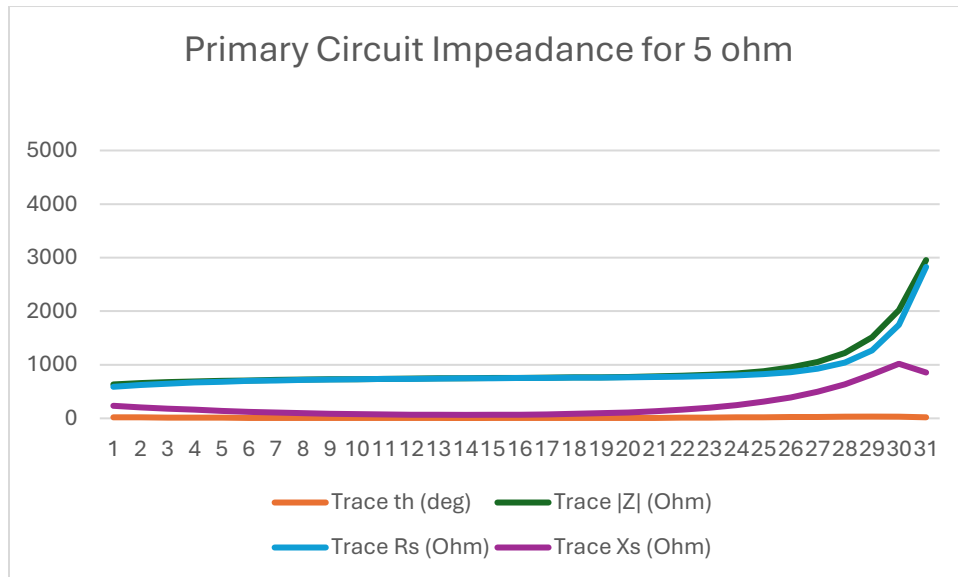


Figure 2.4: Primary Side Impedance for 5 ohm resistor.

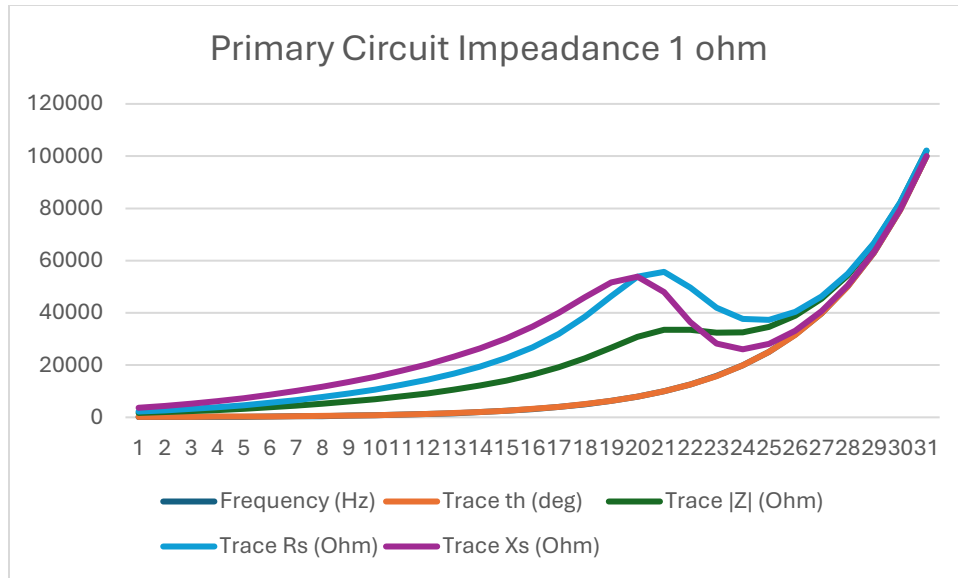


Figure 2.5: Primary Side Impedance for 1 ohm resistor.

Part 3

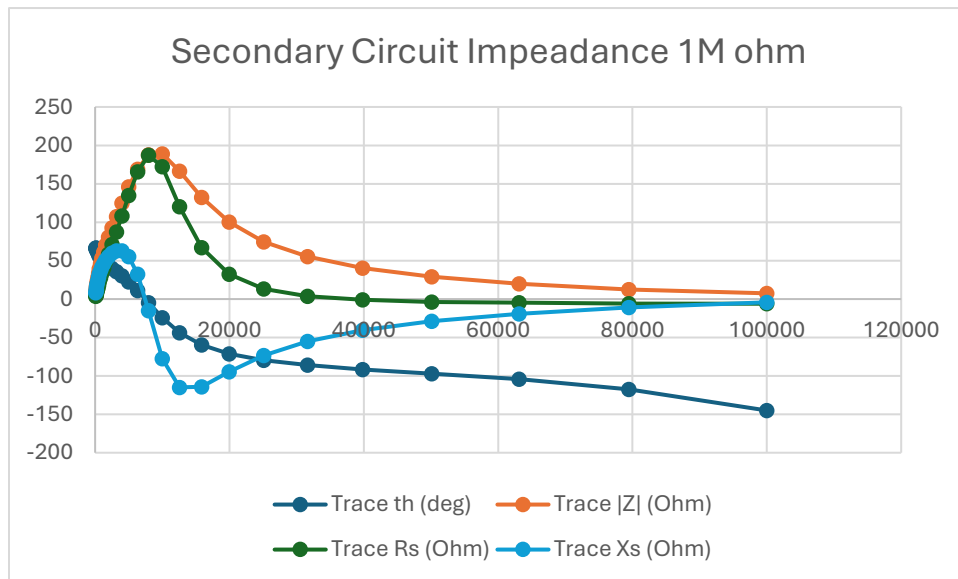


Figure 3.1: Secondary Side Impedance for 1M ohm resistor.

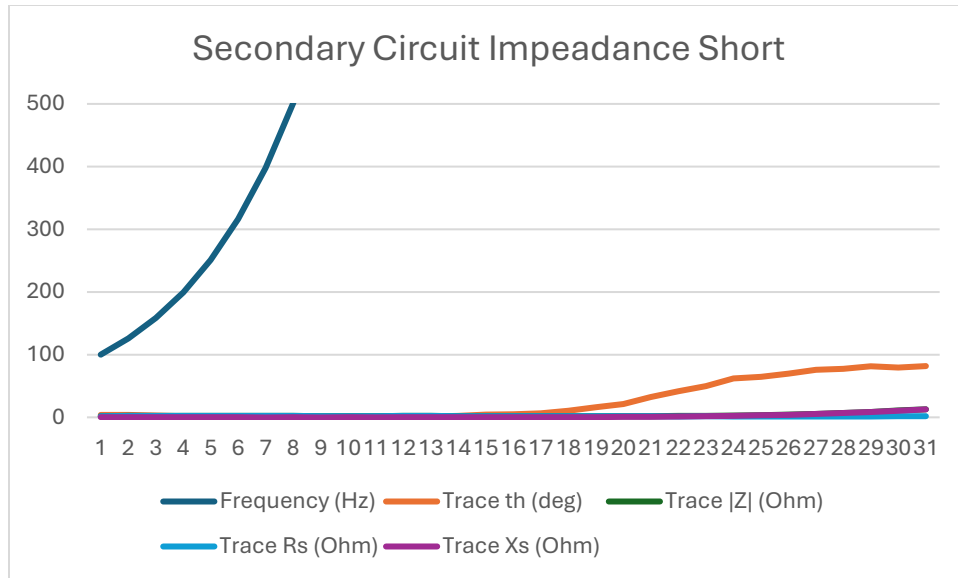


Figure 3.2: Secondary Side Impedance for open circuit.

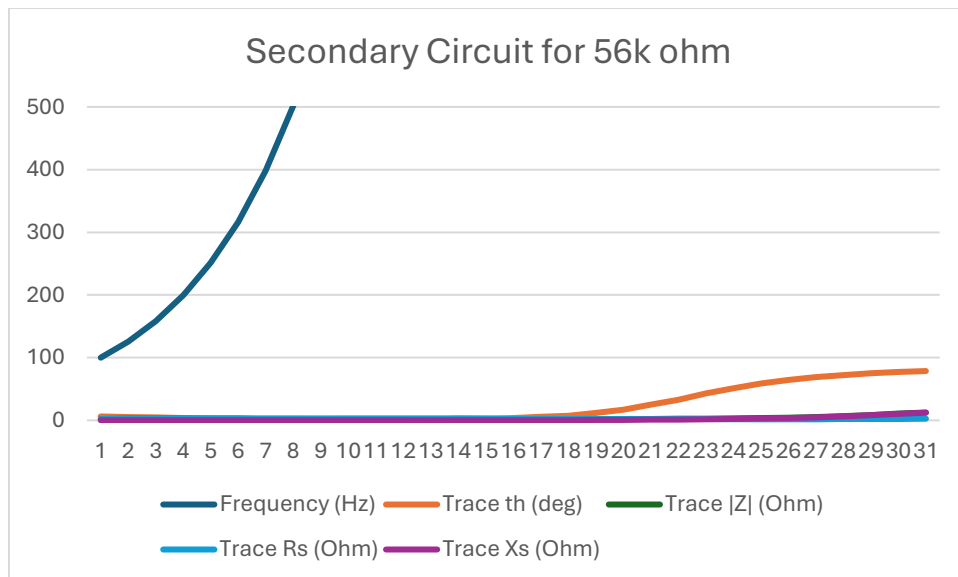


Figure 3.3: Secondary Side Impedance for 56k ohm resistor.

Part 4

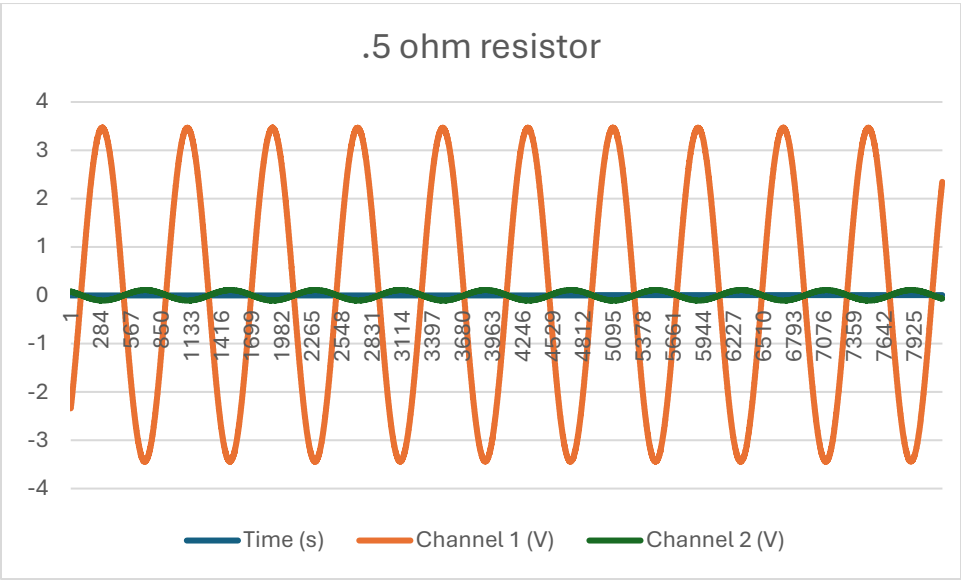


Figure 4.1: Power Transfer of Transformer Circuit with .5 ohm resistor.

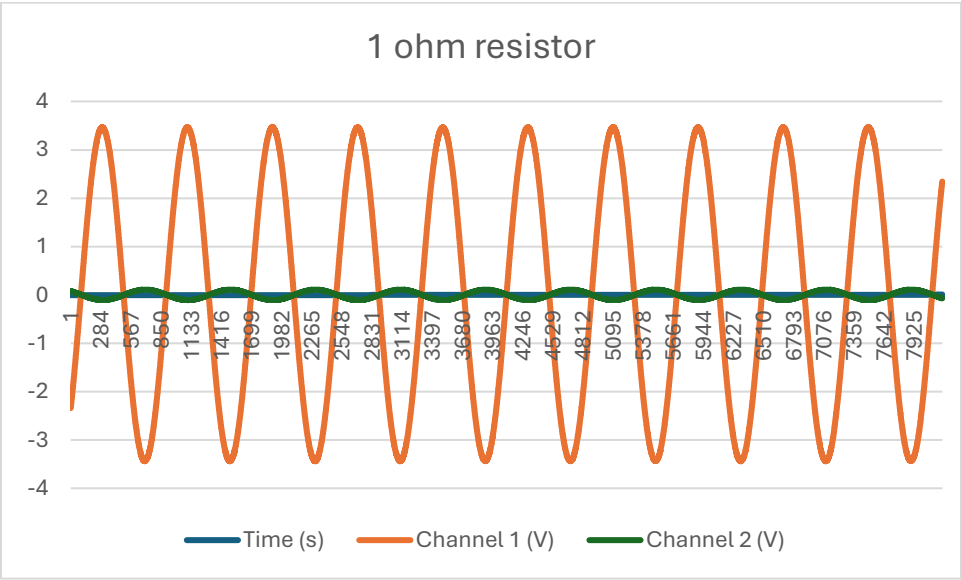


Figure 4.2: Power Transfer of Transformer Circuit with 1 ohm resistor.

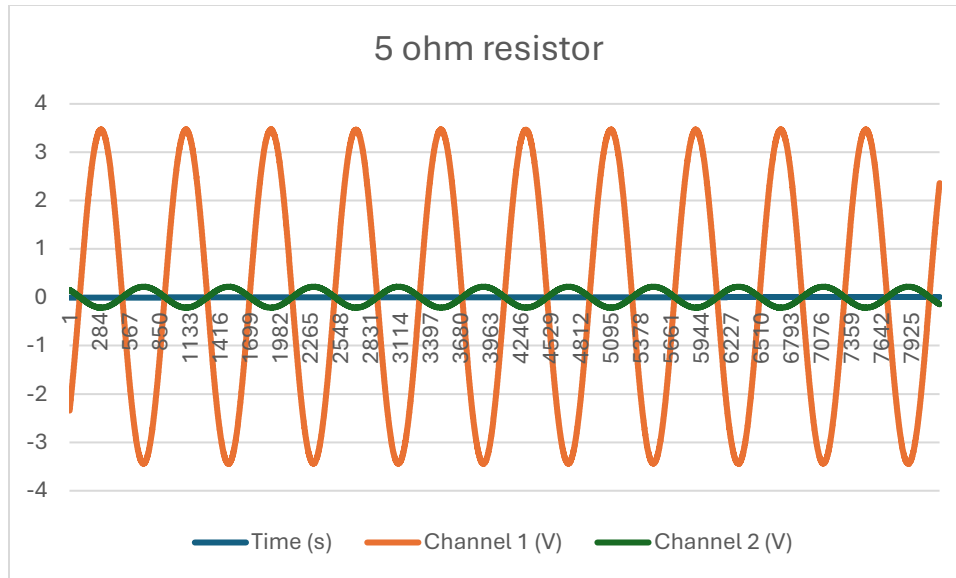


Figure 4.3: Power Transfer of Transformer Circuit with 5 ohm resistor.

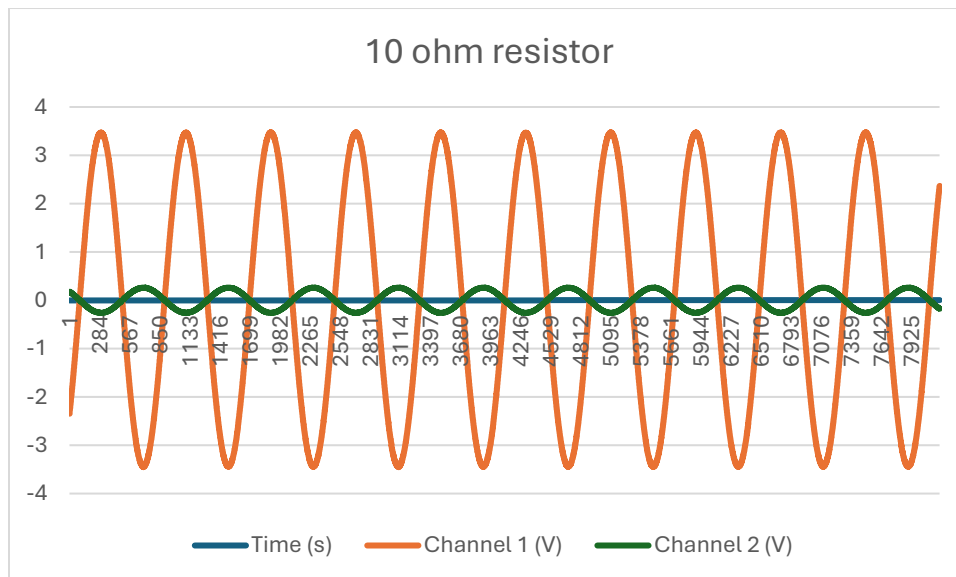


Figure 4.4: Power Transfer of Transformer Circuit with 10 ohm resistor.

Calculations:

1: 10 Ohm: C1: Amplitude: 3.550 V, Mean: 3.4478; C2: Ampl: 259.78 mv, Mean: 259.49 mV

2: 5 Ohm: C1: Amplitude: 3.4478 V, Mean: 3.4468; C2: Ampl: 216.25 mv, Mean: 215.82 mV

3: 1 Ohm: C1: Amplitude: 3.4440 V, Mean: 3.4434; C2: Ampl: 107.62 mv, Mean: 107.48 mV

II. Analysis and Thoughts

Part 1

For this part we measured the transformer under different load conditions the 1M ohm (open circuit), 8.2k ohm, 5-ohm, 1-ohm resistor. We measured the gain of the transformer with the network analyzer as seen in Figure 1.1, 1.3, 1.5, 1.7. In addition, using the oscilloscope to capture the sine waves voltage value at the primary and secondary sides of the transformer as seen in figure 1.2, 1.4, 1.6, 1.8. The data from the network analyzer and the oscilloscope are correlating with each other because the gain from the network analyzer allows us to understand the difference in voltage in the oscilloscope.

The gain on the network analyzer changes for each load condition used we had the smallest gain for the 1M ohm resistor and the largest gain for the 1-ohm resistor. The data in our network analyzer remained more constant throughout our frequency range when the load condition had larger resistance. As you can see in Figure 1.1 and 1.7 there is a drastic difference at the end of the frequency range the gain curves upwards. Therefore, the larger load condition was better higher frequencies.

Viewing our data on the oscilloscope we can see the step-down ratio very clearly and observe the signals are not in phase with each other. In fact, there is a phase shift of 180 degrees. However, the relative phase is the same for all frequency values. After plotting the inverse of the step-down ratio as a function of load impedance we can see the step-down ratio is not constant. Therefore, the transformer is dependent on the load conditions as we saw in our data the ratios changed for each resistor value we used this is confirmed with our data above. There is also a frequency range for where the step-down ratio is constant usually between 1k to 10k hz. These values can vary due to the load conditions and the transformers' properties.

Part 2

In this part we are using the impedance analyzer to measure the impedance on the primary side of the circuit with a frequency function. The primary input impedance was measured across frequencies from 100 Hz to 100 kHz for various secondary loads. With the secondary open-circuited ($1\text{ M}\Omega$), the series resistance (RS) remained low and relatively constant, while the series reactance (XS) increased with frequency, showing the expected inductive behavior. As the secondary load decreased ($8.2\ \Omega$, $5\ \Omega$, $1\ \Omega$, and short circuit), RS increased and XS deviated from a linear trend, due to reflected impedance from the secondary and core losses. With a $0.1\ \mu\text{F}$ capacitive load, the input impedance decreased at higher frequencies, reflecting the reactive nature of the load. Comparing the measured RS to the ideal calculated values, RS was within $\pm 20\%$ over most of the audio range (200 Hz–10 kHz) for moderate to high resistance loads, while low-resistance loads caused larger deviations. Open-circuit measurements also allowed estimation of the primary inductance, which decreased slightly with frequency due to non-ideal core effects.

Part 3

In this part we are using the impedance analyzer to measure the impedance on the secondary side of the circuit with a frequency function. Secondary input impedance measurements followed similar patterns. With an open circuit ($1\text{ M}\Omega$), R_S was low and X_S increased with frequency. Lower-resistance loads (56 k and below) increased R_S and caused X_S to deviate from the ideal trend because of interactions with the primary and non-ideal effects like leakage inductance and core losses. Comparing measured R_S to theoretical values showed agreement within $\pm 20\%$ over the audio range for higher-resistance loads, while lower-resistance loads caused more deviation. Open-circuit secondary measurements allowed calculation of the secondary inductance, which also decreased slightly at higher frequencies. Overall, the transformer behaves approximately as a series RL element, but the frequency range over which it is close to ideal depends on the secondary load.

Part 4

In this part of the lab, we examined how the transformer allows the AD2 to deliver power to low impedance loads without distorting the waveform. Without the transformer, very small load resistances would draw more current than the AD2's hard limits, causing the waveform to clip. By measuring voltages across the primary, secondary, and various load resistances— $10\ \Omega$, $5\ \Omega$, $1\ \Omega$, and $0.5\ \Omega$ —we observed that the transformer stepped down the voltage and increased current to the load, keeping the waveform undistorted. As the load resistance decreased, the voltage across the load also decreased, while the primary voltage increased slightly, reflecting the small output impedance of the AD2. Calculations of power showed that most of the delivered power went to the load, while only a small fraction was lost in the transformer or dissipated by the AD2's output resistance. The output impedance of the AD2 appeared nearly constant across the loads, around $1\ \Omega$, which is consistent with its specifications, while the fraction of total power delivered to the load decreased slightly with very low resistance due to increased losses. Overall, these results demonstrate that the transformer efficiently transfers power from the AD2 to low-resistance loads, allowing high-current delivery without clipping and confirming the transformer's role in safe, effective power scaling.

III. References

None